

Ulf Skutnabba, twitter: @skutis77

Taito K1100319A

Sheet: /
File: risle.kicad_sch

Title: Rainbow Islands Extra

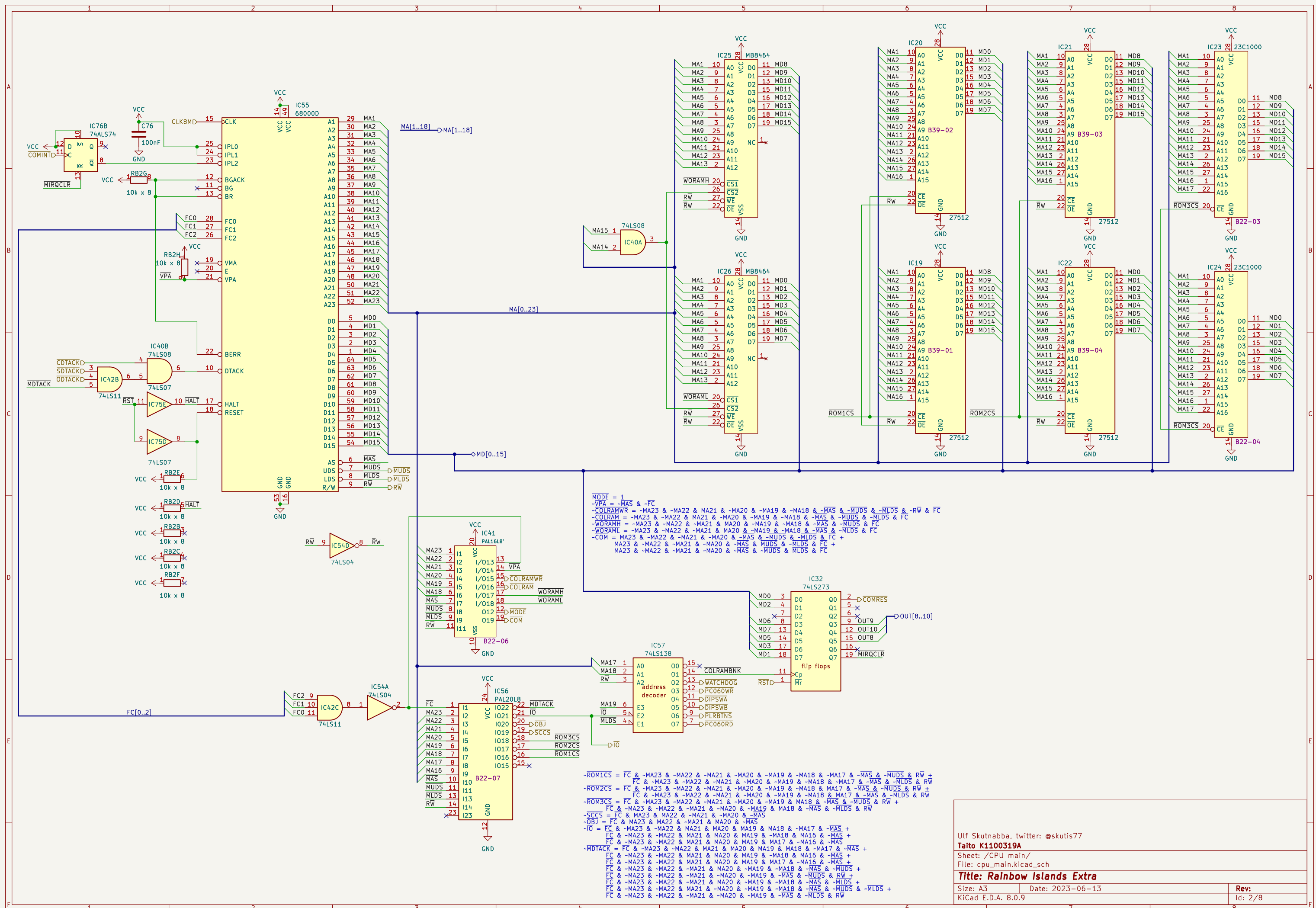
Size: A3	Date: 2023-06-13
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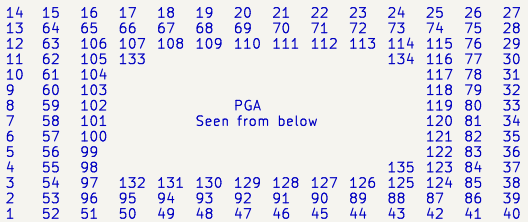
KiCad E.D.A. 8.0.9

Date: 2023-06-13

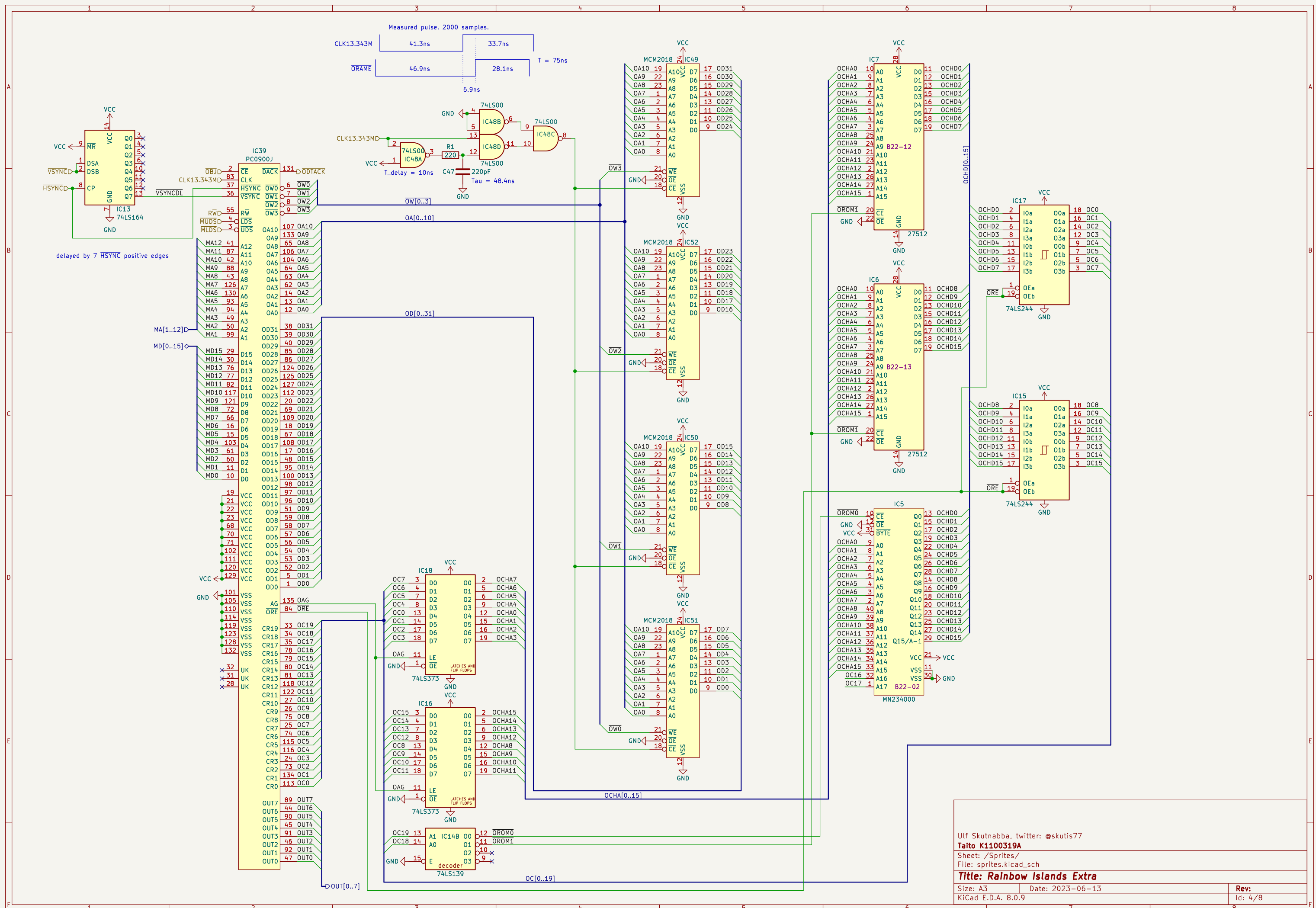
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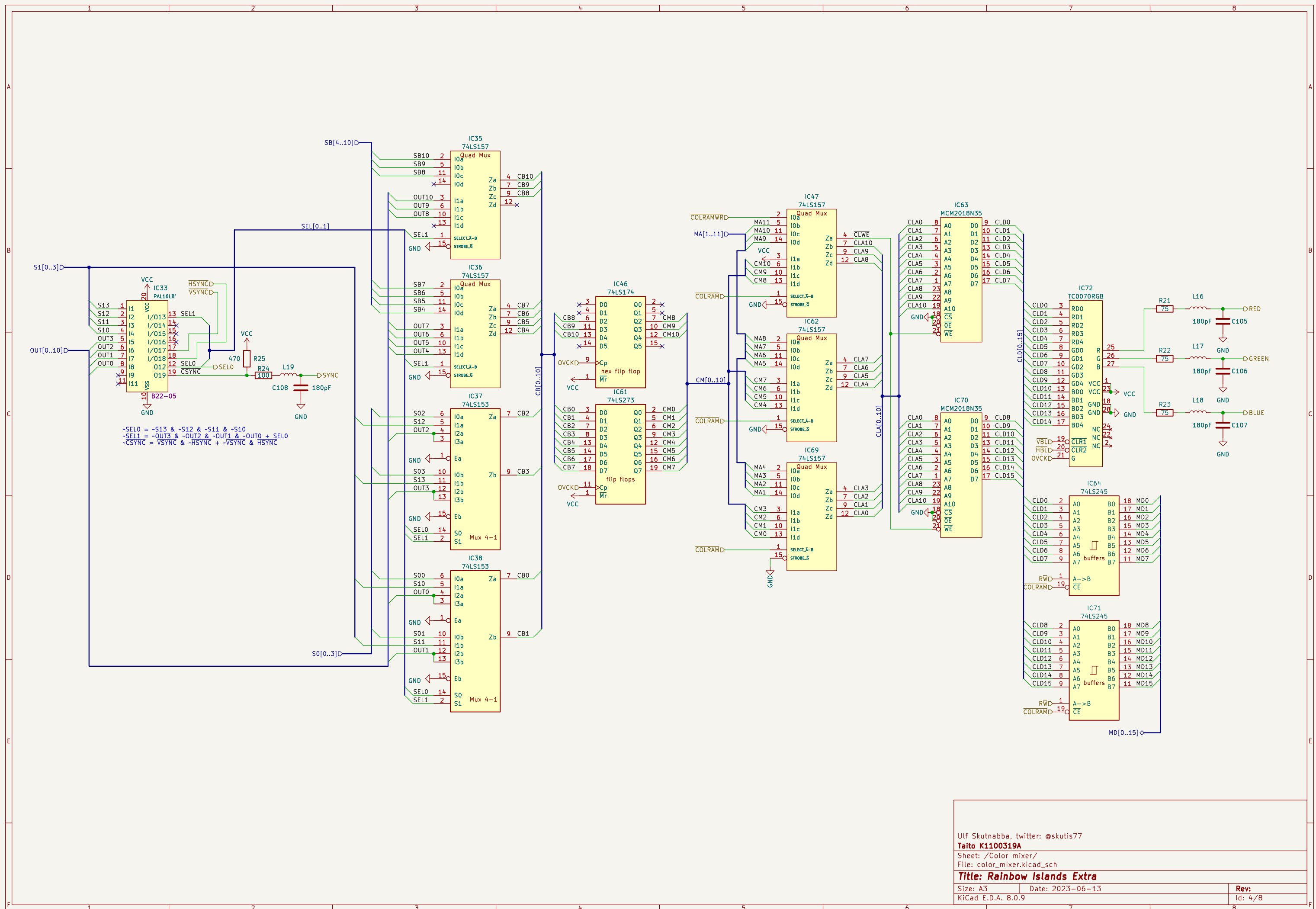
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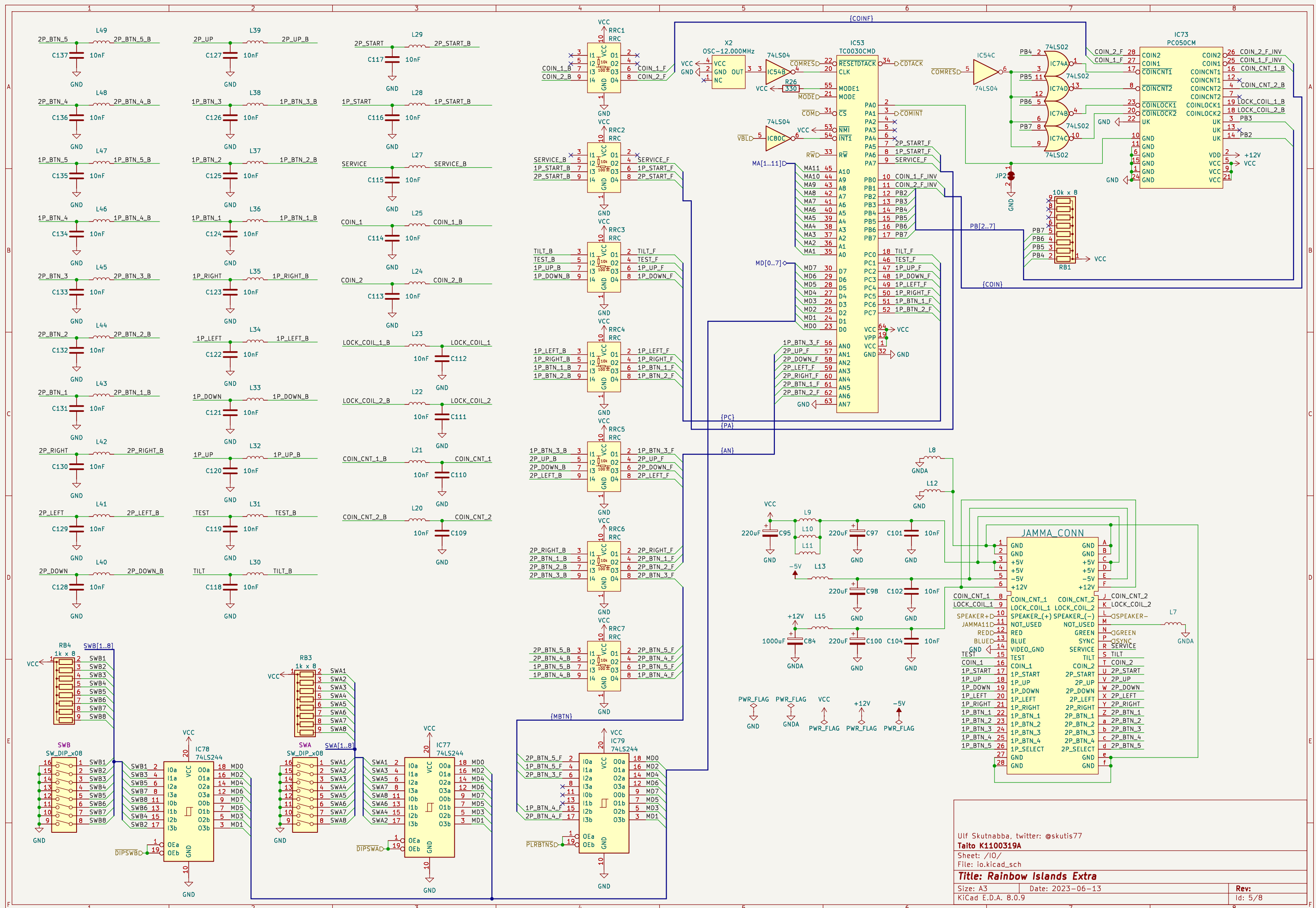




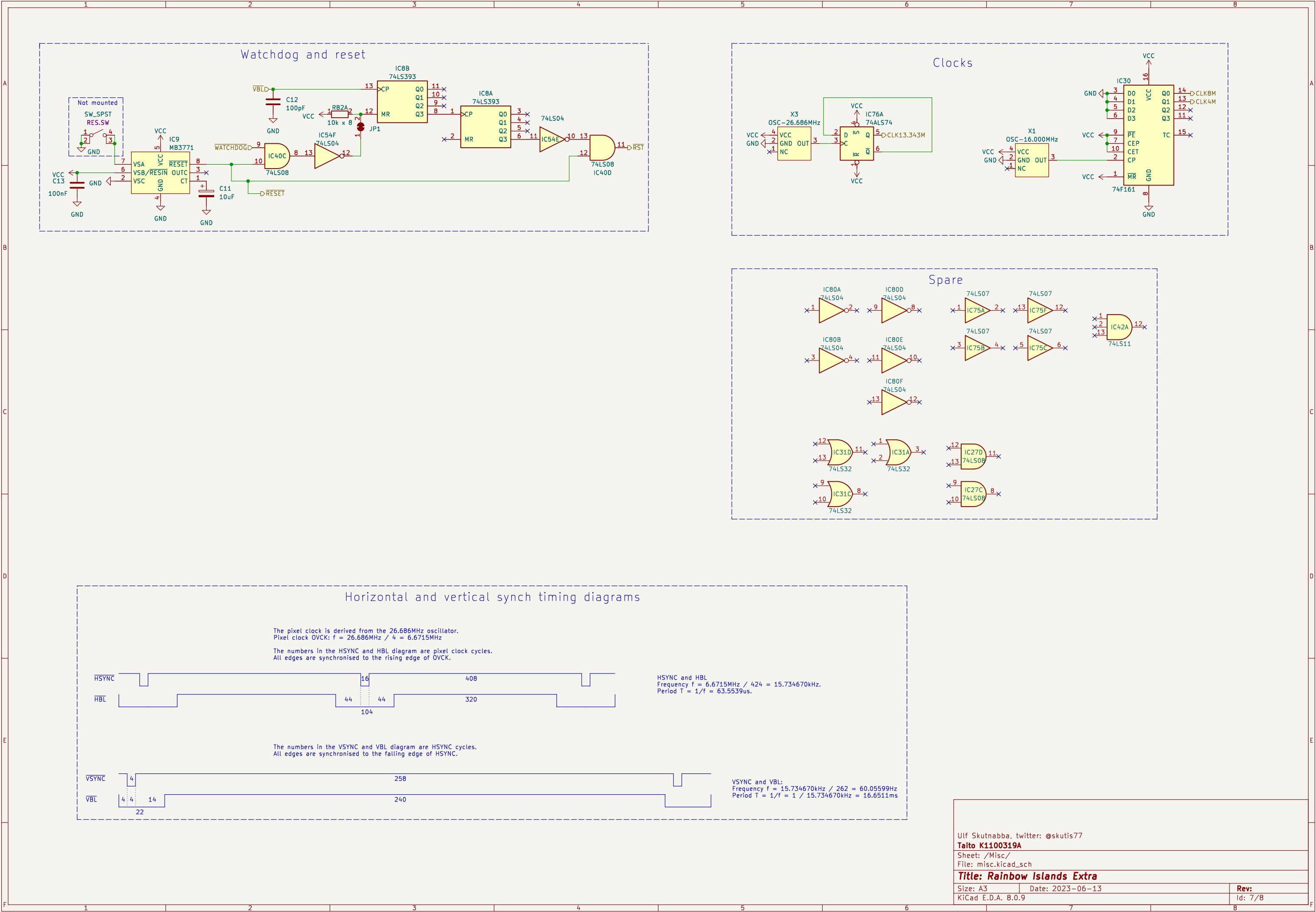
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Horizontal and vertical sync timing diagrams

The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: $f = 26.686\text{MHz} / 4 = 6.6715\text{MHz}$

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.

HSYNC

16

408

44

44

320

104

HBL

HSYNC and HBL
Frequency $f = 6.6715\text{MHz} / 424 = 15.734670\text{kHz}$.
Period $T = 1/f = 63.5539\mu\text{s}$.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

VSYNC

4

258

4

4

14

22

240

VBL

VSYNC and VBL:
Frequency $f = 15.734670\text{kHz} / 262 = 60.05599\text{Hz}$
Period $T = 1/f = 1 / 15.734670\text{kHz} = 16.6511\text{ms}$

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